Category Theory for Mathematical Physics — Exercise Sheet 5

Fusion categories: particles and how they fuse

Variant: Solutions

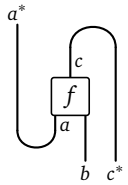
Conventions as in Lecture 5: $N_{ab}^{c} = \dim \mathrm{Hom}(a \otimes b, c)$, the fusion matrix N_a has entries $(N_a)_{cb} = N_{ab}^{c}$, and d_a is the quantum dimension of the canonical spherical structure (Theorem 5.5). Permutations compose right to left, $(gh)(i) = g(h(i))$, and S_3 acts on $\mathbb{C}^3$ by permuting coordinates. The level is an integer $k \geq 1$, the label j of $\mathrm{SU}(2)_k$ is a spin, $0 \leq j \leq k/2$, $\theta = \pi/(k+2)$, and $\varphi = (1+\sqrt{5})/2$. A splitting vertex carries its labels, ψ_c^{ab}; the bare ψ is Ising's fermion. The f_n of Exercise 4 are Fibonacci numbers.

Exercise 1 (Two fusion rings from S_3). (i) In Vec_{S_3} (trivial ω) show that $N_g N_h = N_{gh}$ for all $g, h \in S_3$, deduce associativity of the fusion ring, and show that $N_{(12)}$ and $N_{(23)}$ do not commute.

(ii) Draw the bending picture of Theorem 5.3 that turns $f\colon a \otimes b \to c$ into a morphism $b \otimes c^* \to a^*$. Then check $N_{ab}^{c} = N_{bc^*}^{a^*}$ directly: in Vec_{S_3} for all g, h, k, and in $\mathrm{Rep}(S_3)$ from the fusion rules of Example 5.2.

Solution. (i) The simple objects are the $g \in S_3$, and $g \otimes b = gb$, so $N_{gb}^{c} = 1$ exactly when $c = gb$: N_g is the permutation matrix sending the basis vector b to gb. Then $N_g N_h$ sends b to $g(hb) = (gh)b$, as N_{gh} does. Associativity of the ring, $N_a N_b = \sum_c N_{ab}^{c} N_c$, here reads $N_g N_h = N_{gh}$; it holds because $(gh)b = g(hb)$, the associativity of the group. With $(12)(23) = (123)$ (first (23), then (12): $3 \mapsto 2 \mapsto 1$, $1 \mapsto 2$, $2 \mapsto 3$) and $(23)(12) = (132)$, $N_{(12)} N_{(23)} = N_{(123)} \neq N_{(132)} = N_{(23)} N_{(12)}$: the ring is not commutative.

(ii) The picture is the one in the proof of Theorem 5.3: the c output of f is bent down to the right into an input c^* by a cap, and the a input is bent up to the left into an output a^* by a cup,



and bending back undoes it (the snakes), so $\dim \mathrm{Hom}(a \otimes b, c) = \dim \mathrm{Hom}(b \otimes c^*, a^*)$. In Vec_{S_3}, $g^* = g^{-1}$, so $N_{hk^{-1}}^{g^{-1}} = 1$ exactly when $hk^{-1} = g^{-1}$, that is $k = gh$: the same condition as $N_{gh}^{k} = 1$. In $\mathrm{Rep}(S_3)$ all three simple objects are self-dual, and the rules of Example 5.2 give $N_{ab}^{c} = 1$ for the five multisets $\{1,1,1\}$, $\{1,\mathrm{sgn},\mathrm{sgn}\}$, $\{1,\mathrm{std},\mathrm{std}\}$, $\{\mathrm{sgn},\mathrm{std},\mathrm{std}\}$, $\{\mathrm{std},\mathrm{std},\mathrm{std}\}$ (as multisets) and 0 otherwise (for instance $\mathrm{sgn} \otimes \mathrm{std} \cong \mathrm{std}$ gives $N_{\mathrm{sgn}\,\mathrm{std}}^{\mathrm{std}} = 1$, and $\mathrm{std} \otimes \mathrm{std}$ contains each simple object once). This list depends only on the multiset $\{a,b,c\}$, so it is unchanged by $(a,b,c) \mapsto (b,c,a)$, which is the identity $N_{ab}^{c} = N_{bc}^{a}$ for self-dual objects.

Exercise 2 (Quantum dimensions of $\mathrm{SU}(2)_k$). Let $d_j = \sin\big((2j+1)\theta\big)/\sin\theta$ for $j = -\frac{1}{2}, 0, \frac{1}{2}, \ldots, \frac{k+1}{2}$.

(i) Show that $d_{-1/2} = d_{(k+1)/2} = 0$ and that

$$d_{1/2}\, d_j = d_{j-1/2} + d_{j+1/2} \qquad \big(0 \leq j \leq \tfrac{k}{2}\big),$$

and explain why this matches the truncated rule for $\frac{1}{2} \otimes j$, including at $j = \frac{k}{2}$.

(ii) Show that every $[j]$ in the fusion ring is a polynomial in $[\frac{1}{2}]$ with integer coefficients. Show that an additive map f from the fusion ring to $\mathbb{R}$ with $f([0]) = 1$ and $f([\frac{1}{2}]z) = f([\frac{1}{2}])f(z)$ for every z is multiplicative. Conclude with Theorem 5.4 that $d_j = \mathrm{FPdim}(j)$.

(iii) List the d_j for $k = 2, 3, 4$.

Solution. (i) $d_{-1/2} = \sin 0/\sin\theta = 0$ and $d_{(k+1)/2} = \sin\big((k+2)\theta\big)/\sin\theta = \sin\pi/\sin\theta = 0$. Also $d_{1/2} = \sin 2\theta/\sin\theta = 2\cos\theta$, and with $x = (2j+1)\theta$, $2\cos\theta\,\sin x = \sin(x+\theta) + \sin(x-\theta) = \sin\big((2j+2)\theta\big) + \sin(2j\theta)$; dividing by $\sin\theta$ gives the recursion. The truncated rule gives $\frac{1}{2} \otimes j \cong (j - \frac{1}{2}) \oplus (j + \frac{1}{2})$, where $j - \frac{1}{2}$ is absent for $j = 0$ and $j + \frac{1}{2}$ is absent exactly when $j + \frac{1}{2} > k - \frac{1}{2} - j$, that is $j = \frac{k}{2}$. The recursion has the same form, and the two missing terms are $d_{-1/2} = 0$ and $d_{(k+1)/2} = 0$.

(ii) For $0 \le j \le \frac{k-1}{2}$ the rule reads $[j+\frac{1}{2}] = [\frac{1}{2}][j] - [j-\frac{1}{2}]$ (with $[-\frac{1}{2}] = 0$), so by induction on $2j$ each $[j] = P_j([\frac{1}{2}])$, with $P_0 = 1$, $P_{1/2} = x$ and $P_{j+1/2} = xP_j - P_{j-1/2}$, integer polynomials. Now let f be as stated. By induction, $f([\frac{1}{2}]^n z) = f([\frac{1}{2}])\,f([\frac{1}{2}]^{n-1}z) = f([\frac{1}{2}])^n f(z)$. For $x = P([\frac{1}{2}]) = \sum_n a_n[\frac{1}{2}]^n$, additivity gives $f(xz) = \sum_n a_n f([\frac{1}{2}])^n f(z) = P(f([\frac{1}{2}]))\,f(z)$, and with $z = [0]$, $f(x) = P(f([\frac{1}{2}]))$; so $f(xz) = f(x)f(z)$ for all x, z. The map $f([j]) = d_j$ has $f([0]) = d_0 = 1$, and $f([\frac{1}{2}][j]) = d_{j-1/2} + d_{j+1/2} = d_{1/2}d_j$ by (i) (at $j = \frac{k}{2}$ the product is $[\frac{k-1}{2}]$, and $d_{(k+1)/2} = 0$); by additivity this holds for every z. So d is multiplicative. It is positive: for $0 \le j \le \frac{k}{2}$, $0 < (2j+1)\theta \le (k+1)\theta < \pi$. Every simple object is self-dual, so by the uniqueness part of Theorem 5.4 (a positive multiplicative function with value 1 on the unit is FPdim), $d_j = \mathrm{FPdim}(j)$.

(iii) $k = 2$, $\theta = \pi/4$: $1, \sqrt{2}, 1$. $k = 3$, $\theta = \pi/5$: $d_{1/2} = 2\cos(\pi/5) = \varphi$, $d_1 = \sin(3\pi/5)/\sin(\pi/5) = \varphi$ (as $\sin(3\pi/5) = \sin(2\pi/5)$), and $d_{3/2} = 1$. $k = 4$, $\theta = \pi/6$: $d_{1/2} = 2\cos(\pi/6) = \sqrt{3}$, $d_1 = \sin(\pi/2)/\sin(\pi/6) = 2$, then $\sqrt{3}$, 1. The values for $k = 2$ are Ising's, as the rings agree; for $k = 3$, 0 and 1 have Fibonacci's.

Exercise 3 (Vertices for std ⊗ std**).** Realise std on $\{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ with the orthonormal basis $v_1 = (e_1 - e_2)/\sqrt{2}$, $v_2 = (e_1 + e_2 - 2e_3)/\sqrt{6}$, and write $v_{ab} = v_a \otimes v_b$. Let

$$w_{\mathbf{1}}(1) = \tfrac{1}{\sqrt{2}}(v_{11} + v_{22}), \qquad w_{\mathrm{sgn}}(1) = \tfrac{1}{\sqrt{2}}(v_{12} - v_{21}),$$
$$w_{\mathrm{std}}(v_1) = \tfrac{1}{\sqrt{2}}(v_{12} + v_{21}), \qquad w_{\mathrm{std}}(v_2) = \tfrac{1}{\sqrt{2}}(v_{11} - v_{22}).$$

(i) Write the matrices of (12) and (23) on std, and show that each w_c is an isometric intertwiner $c \to \mathrm{std} \otimes \mathrm{std}$.

(ii) With the splitting vertices $\psi_c^{\mathrm{std\,std}} = (4/d_c)^{1/4}\, w_c$, check the bubble Equation (5.1) for all c, c', and the theta graphs Equation (5.4).

(iii) Draw the completeness relation Equation (5.3) for std ⊗ std, and check it as an identity of 4×4 matrices.

Solution. (i) (12) swaps e_1, e_2: $v_1 \mapsto -v_1$, $v_2 \mapsto v_2$. (23) swaps e_2, e_3: $v_1 \mapsto (e_1 - e_3)/\sqrt{2} = \frac{1}{2}v_1 + \frac{\sqrt{3}}{2}v_2$ and $v_2 \mapsto (e_1 - 2e_2 + e_3)/\sqrt{6} = \frac{\sqrt{3}}{2}v_1 - \frac{1}{2}v_2$ (take inner products with v_1, v_2). Both matrices are real orthogonal reflections, of determinant -1, as sgn of a transposition requires. The vectors $\frac{1}{\sqrt{2}}(v_{11} + v_{22})$, $\frac{1}{\sqrt{2}}(v_{12} - v_{21})$, $\frac{1}{\sqrt{2}}(v_{12} + v_{21})$, $\frac{1}{\sqrt{2}}(v_{11} - v_{22})$ are orthonormal, so each w_c is an isometry and their images are orthogonal. Intertwining, with $\rho = \rho_{\mathrm{std}}(g)$ real orthogonal: $\sum_i \rho v_i \otimes \rho v_i = \sum_i v_i \otimes v_i$, so $w_{\mathbf{1}}$ is invariant. For w_{sgn}, $\rho v_1 \otimes \rho v_2 - \rho v_2 \otimes \rho v_1 = \det(\rho)(v_{12} - v_{21})$, and $\det\rho = \mathrm{sgn}(g)$. For w_{std} and $g = (12)$: $w_{\mathrm{std}}(\rho v_1) = -w_{\mathrm{std}}(v_1)$ and $(\rho \otimes \rho)w_{\mathrm{std}}(v_1) = \frac{1}{\sqrt{2}}(-v_{12} - v_{21})$; $w_{\mathrm{std}}(\rho v_2) = w_{\mathrm{std}}(v_2)$ and $(\rho \otimes \rho)w_{\mathrm{std}}(v_2) = w_{\mathrm{std}}(v_2)$. For

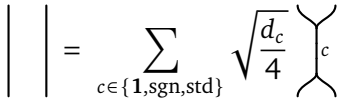
$g = (23)$, write $\rho v_1 = \frac{1}{2}v_1 + sv_2$, $\rho v_2 = sv_1 - \frac{1}{2}v_2$ with $s = \frac{\sqrt{3}}{2}$:

$$\begin{aligned}(\rho\otimes\rho)(v_{12}+v_{21}) &= s(v_{11}-v_{22}) + \left(s^2-\tfrac{1}{4}\right)(v_{12}+v_{21}) \\ &= \tfrac{1}{2}(v_{12}+v_{21}) + s(v_{11}-v_{22}),\end{aligned}$$

which is $\sqrt{2}\,w_{\mathrm{std}}(\frac{1}{2}v_1 + sv_2) = \sqrt{2}\,w_{\mathrm{std}}(\rho v_1)$; similarly $(\rho\otimes\rho)(v_{11}-v_{22}) = s(v_{12}+v_{21}) - \frac{1}{2}(v_{11}-v_{22}) = \sqrt{2}\,w_{\mathrm{std}}(\rho v_2)$. As (12) and (23) generate S_3, each w_c is an intertwiner.

(ii) $(\psi_c^{\mathrm{std}\,\mathrm{std}})^\dagger \psi_{c'}^{\mathrm{std}\,\mathrm{std}} = (4/d_c)^{1/4}(4/d_{c'})^{1/4} w_c^\dagger w_{c'}$, which is 0 for $c \neq c'$ (orthogonal images) and $(4/d_c)^{1/2}\,\mathrm{id}_c = \sqrt{d_{\mathrm{std}}d_{\mathrm{std}}/d_c}\;\mathrm{id}_c$ for $c = c'$: Equation (5.1), with values 2, 2, $\sqrt{2}$ for $c = \mathbf{1}, \mathrm{sgn}, \mathrm{std}$. The vacuum bubble is $2 = d_{\mathrm{std}}$, and $\psi_{\mathbf{1}}^{\mathrm{std}\,\mathrm{std}}(1) = v_{11} + v_{22}$ is the cup of Equation (5.2), as in the lecture. Closing the c strand takes the trace over c: the theta graphs are 2, 2 and $\sqrt{2}\cdot 2 = 2\sqrt{2}$, which are $\sqrt{2\cdot 2\cdot d_c}$.

(iii) The picture is Equation (5.3) with $a = b = \mathrm{std}$ and three terms:



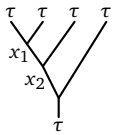
all four outer strands labelled std. The term for c is $\sqrt{d_c/4}\,\psi_c^{\mathrm{std}\,\mathrm{std}}(\psi_c^{\mathrm{std}\,\mathrm{std}})^\dagger = w_c w_c^\dagger$, the orthogonal projection onto the image of w_c. The four vectors of (i) are an orthonormal basis of $\mathbb{C}^2\otimes\mathbb{C}^2$, so the projections onto their spans, of ranks 1, 1, 2, sum to I_4.

Exercise 4 (How fusion spaces grow). (i) Draw the fusion trees (Theorem 5.6) with four leaves τ and root τ in the Fibonacci category.

(ii) Let g_n be the vector $(\dim V_1^{\tau\cdots\tau}, \dim V_\tau^{\tau\cdots\tau})$ for $n \geq 1$ leaves τ. Show that $g_n = N_\tau^{n-1} g_1$ with $g_1 = (0,1)$, and hence $g_n = (f_{n-1}, f_n)$ with $f_0 = 0$, $f_1 = 1$, $f_{n+1} = f_n + f_{n-1}$. Prove Binet's formula $f_n = \left(\varphi^n - (-1/\varphi)^n\right)/\sqrt{5}$, and deduce that $f_{n+1}/f_n \to \varphi = d_\tau$. How many states do 20 τ's with total charge τ have?

(iii) In the Ising category, show that for $m \geq 1$, $2m$ copies of σ have 2^{m-1} states with total charge 1 and 2^{m-1} with total charge ψ.

Solution. (i) The inner labels are $x_1 \in \{1, \tau\}$, a channel of $\tau\otimes\tau$, and x_2, a channel of $x_1\otimes\tau$; both $1\otimes\tau$ and $\tau\otimes\tau$ contain τ, so the root puts no further condition. $x_1 = 1$ forces $x_2 = \tau$, and $x_1 = \tau$ allows $x_2 \in \{1, \tau\}$. So the trees are



$$(x_1, x_2) \in \{(1,\tau), (\tau, 1), (\tau,\tau)\}.$$

(ii) By Theorem 5.6, $\dim V_c^{\tau\cdots\tau}$ with $n+1$ leaves is $\sum_x N_{x\tau}^c \dim V_x^{\tau\cdots\tau}$ with n leaves: $g_{n+1} = N_\tau g_n$, as $(N_\tau)_{cx} = N_{\tau x}^c = N_{x\tau}^c$. With $N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $(u, v) \mapsto (v, u+v)$, so from $g_1 = (0,1) = (f_0, f_1)$ we get $g_n = (f_{n-1}, f_n)$. The sequence $\left(\varphi^n - (-1/\varphi)^n\right)/\sqrt{5}$ is $0, 1$ at $n = 0, 1$ and obeys $f_{n+1} = f_n + f_{n-1}$, since both φ and $-1/\varphi$ satisfy $x^2 = x + 1$; so it is f_n. As $|-1/\varphi| < 1$, $f_n \approx \varphi^n/\sqrt{5}$ and $f_{n+1}/f_n \to \varphi$. For $n = 20$: $f_{20} = 6765$.

(iii) In the basis $1, \sigma, \psi$, N_σ swaps the parity: it maps the span of $1, \psi$ to that of σ and back, and $N_\sigma^2(u, 0, v) = (u+v, 0, u+v)$. With $g_1 = (0,1,0)$, $g_2 = N_\sigma g_1 = (1,0,1)$, and then $g_{2m} = N_\sigma^{2(m-1)} g_2 = (2^{m-1}, 0, 2^{m-1})$ by induction. So $2m$ σ's carry 2^{m-1} states of each total charge, about $(\sqrt{2})^{2m}/2$: growth like d_σ^n again.

Exercise 5 (D_8 and Q_8). Let G be D_8 or Q_8, given by the 2×2 matrices of Sheet 4, acting on $V=\mathbb{C}^2$. You may use, from *Symmetries* (cited in Lecture 5), that irreducible characters are orthonormal for $\langle\chi,\chi'\rangle = \frac{1}{|G|}\sum_g \chi(g)\overline{\chi'(g)}$, that the irreducible representations have $\sum \dim^2 = |G|$, and that $N_{ab}^c = \langle \chi_a\chi_b, \chi_c\rangle$; and that in both groups $\{\pm 1\}$ is a normal subgroup with quotient $\mathbb{Z}_2\times\mathbb{Z}_2$.

(i) Show that G has exactly five irreducible representations: four of dimension 1, pulled back from $\mathbb{Z}_2\times\mathbb{Z}_2$ (call them $\mathbf{1},\chi_1,\chi_2,\chi_3$, with $\chi_1\chi_2=\chi_3$), and V.

(ii) Compute all fusion rules of $\mathrm{Rep}(G)$, and show that $\mathrm{Rep}(D_8)$ and $\mathrm{Rep}(Q_8)$ have the same fusion ring.

(iii) Give the FPdim's and the global dimension.

Solution. (i) The four characters of $\mathbb{Z}_2\times\mathbb{Z}_2$, composed with $G\to G/\{\pm1\}\cong\mathbb{Z}_2\times\mathbb{Z}_2$, are four distinct 1-dimensional representations, irreducible. $\langle\chi_V,\chi_V\rangle = \frac{1}{8}(2^2+(-2)^2)=1$, so V is irreducible, and it is not 1-dimensional. The squares of the dimensions already add up to $4+4=8=|G|$, so there are no others. The labels χ_1,χ_2 can be chosen with $\chi_1\chi_2=\chi_3$, since the characters of $\mathbb{Z}_2\times\mathbb{Z}_2$ form a group $\mathbb{Z}_2\times\mathbb{Z}_2$.

(ii) Characters multiply under $\otimes$. The χ's multiply as the group $\mathbb{Z}_2\times\mathbb{Z}_2$: $\chi_i\otimes\chi_i\cong\mathbf{1}$ and $\chi_1\otimes\chi_2\cong\chi_3$, and so on. Since χ_V vanishes off $\{\pm1\}$ and $\chi_i(\pm1)=1$, $\chi_i\chi_V=\chi_V$: $\chi_i\otimes V\cong V$. And χ_V^2 is 4 on $\{\pm1\}$ and 0 elsewhere, which is $\mathbf{1}+\chi_1+\chi_2+\chi_3$ (the sum of the characters of $\mathbb{Z}_2\times\mathbb{Z}_2$ is 4 at the identity and 0 elsewhere, pulled back): $V\otimes V\cong\mathbf{1}\oplus\chi_1\oplus\chi_2\oplus\chi_3$. None of this used which group G is, so the bijection $\mathbf{1},\chi_1,\chi_2,\chi_3,V\mapsto\mathbf{1},\chi_1,\chi_2,\chi_3,V$ matches the two fusion rings. The categories are nevertheless not equivalent (Calaque–Etingof, cited in Lecture 5); Sheet 4's indicators, $\kappa_V=+1$ for D_8 and -1 for Q_8, show one difference in their pivotal data.

(iii) The fusion matrices of χ_i are permutation matrices, with eigenvalue 1 on the positive vector $(1,1,1,1,2)$, and N_V maps it to $(2,2,2,2,4)$: so FPdim $=1,1,1,1,2$, the dimensions, as in $\mathrm{Rep}(G)$ always. The global dimension is $4\cdot1+2^2=8=|G|$.

Exercise 6 ⋆. *Global dimensions of the four families.* Recall $\dim C=\sum_a d_a^2$ with the canonical $d_a=\mathrm{FPdim}(a)$.

(i) Show that $\dim\mathrm{Vec}_G=\dim\mathrm{Rep}(G)=|G|$, $\dim\mathrm{Fib}=2+\varphi$ and $\dim\mathrm{Ising}=4$.

(ii) Show that $\sum_{m=1}^{k+1}\sin^2(m\theta)=\frac{k+2}{2}$, and deduce

$$\dim \mathrm{SU}(2)_k=\frac{k+2}{2\sin^2\theta}.$$

Evaluate it for $k=2,3,4$.

(iii) Show that for every k the object $\frac{k}{2}$ is invertible, with $\frac{k}{2}\otimes j\cong\frac{k}{2}-j$, and deduce $d_{k/2-j}=d_j$.

(iv) Let k be odd. Show that the integer spins span a fusion subring, that every simple object is $y\otimes x$ for exactly one $y\in\{0,\frac{k}{2}\}$ and one integer spin x, and that

$$\dim \mathrm{SU}(2)_k=2\sum_{j\in\mathbb{Z}} d_j^2.$$

What is the integer-spin subring for $k=3$? Why does the argument fail for even k? Show that the identity holds for even k all the same, with an alternating version of the sum in (ii).

Solution. (i) In Vec_G every simple object g is invertible, N_g is a permutation matrix and $\mathrm{FPdim}(g)=1$; there are $|G|$ of them. In $\mathrm{Rep}(G)$, FPdim is the dimension (Lecture 5), and $\sum_V\dim(V)^2=|G|$ (*Symmetries*). Fibonacci: $1+\varphi^2=1+(\varphi+1)=2+\varphi$. Ising: $1+(\sqrt{2})^2+1=4$.

(ii) $\sum_{m=1}^{k+1}\sin^2(m\theta)=\frac12\sum_{m=1}^{k+1}\big(1-\cos(2m\theta)\big)$. The numbers $e^{2im\theta}$, $m=0,\dots,k+1$, are the $(k+2)$-th roots of unity, which sum to 0; so $\sum_{m=1}^{k+1}\cos(2m\theta)=-\cos 0=-1$, and the sum is $\frac12\big((k+1)+1\big)=\frac{k+2}{2}$. With $m=2j+1$, running over $1,\dots,k+1$ as j runs over $0,\dots,\frac{k}{2}$, Exercise 2 gives $\dim \mathrm{SU}(2)_k=\sum_j d_j^2=\frac{1}{\sin^2\theta}\cdot\frac{k+2}{2}$. For $k=2$: $4/(2\cdot\frac12)=4$, Ising's value. For $k=4$: $6/(2\cdot\frac14)=12$. For $k=3$, from Exercise 2(iii), $1+\varphi^2+\varphi^2+1=2(2+\varphi)$.

(iii) In $\frac{k}{2}\otimes j$ the channels J satisfy $J\ge\big|\frac{k}{2}-j\big|=\frac{k}{2}-j$ and $J\le\min(\frac{k}{2}+j,\ k-\frac{k}{2}-j)=\frac{k}{2}-j$, so $\frac{k}{2}\otimes j\cong\frac{k}{2}-j$. In particular $\frac{k}{2}\otimes\frac{k}{2}\cong 0$: $\frac{k}{2}$ is invertible, its own inverse. Its dimension is $d_{k/2}=\sin\big((k+1)\theta\big)/\sin\theta=\sin(\pi-\theta)/\sin\theta=1$, and multiplicativity (Exercise 2) gives $d_{k/2-j}=d_{k/2}d_j=d_j$.

(iv) In the truncated rule every channel of $j_1\otimes j_2$ differs from j_1+j_2 by an integer, so integer spins fuse into integer spins: they span a subring, containing 0. For odd k, $\frac{k}{2}$ is a half-integer, so $x\mapsto\frac{k}{2}\otimes x=\frac{k}{2}-x$ is a bijection from the integer spins onto the half-integer ones (its inverse is the same map, by (iii)). Hence every simple object j is either $0\otimes x=x$ with x an integer spin, or $\frac{k}{2}\otimes x$ with $x=\frac{k}{2}-j$, and not both, since the two kinds have different parity; and x is determined by the object. Using (iii),

$$\dim \mathrm{SU}(2)_k=\sum_{x\in\mathbb{Z}}d_x^2+\sum_{x\in\mathbb{Z}}d_{k/2-x}^2=2\sum_{x\in\mathbb{Z}}d_x^2.$$

For $k=3$ the integer spins are $0,1$ with $1\otimes 1\cong 0\oplus 1$: the Fibonacci ring, with $d_1=\varphi$, and $\dim \mathrm{SU}(2)_3=2(1+\varphi^2)=2\dim \mathrm{Fib}$, as in (ii). For even k, $\frac{k}{2}$ is itself an integer spin, so tensoring with it stays among the integer spins and the half-integer spins are not reached. The identity holds nevertheless. Integer j have odd $m=2j+1$, so it says $\sum_{m=1}^{k+1}(-1)^m\sin^2(m\theta)=0$. Now $\sum_{m=1}^{k+1}(-1)^m=-1$ for even k, and $\sum_{m=0}^{k+1}(-e^{2i\theta})^m=\big(1-(-e^{2i\theta})^{k+2}\big)/(1+e^{2i\theta})=0$, since $(-1)^{k+2}e^{2i\pi}=1$ and $e^{2i\theta}\ne-1$; so $\sum_{m=1}^{k+1}(-1)^m\cos(2m\theta)=-1$, and $\sum_{m=1}^{k+1}(-1)^m\sin^2(m\theta)=\frac12(-1)-\frac12(-1)=0$.